Predictive Neuromorphic Control of a Scanning Mirror System

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***Abstract*— Organizing control around events can reduce sensing and actuation demands while directly linking control decisions to the occurrence and timing of desired system behaviors. This paper proposes a novel event-centric model predictive control (MPC) framework that unifies event sensing, event prediction, and event regulation. Events define the feedback information, prediction horizon, and event-timing objectives, placing them at the center of predictive control design. The framework is developed for a scanning mirror system equipped with a baseline neuromorphic controller composed of spiking neurons that supports sparse actuation. The control objective is to achieve prescribed zero-crossing times and a desired scanning range while minimizing the energy exchange between the controller and the system. Sampling-based algorithms are presented to obtain suboptimal solutions of the event-centric MPC optimization problem without requiring an explicit event-to-event prediction model. By grounding predictive optimization in events, the proposed framework opens a new direction for MPC design and its integration with neuromorphic control.**

I. Introduction

Neuromorphic control exploits the dynamics of excitable neuronal circuits to regulate behavior through sensing and actuation events [1]. It is particularly relevant to rhythmic systems, where control relies on coordinating recurring events, with applications including pendulum control [2] and soft robotics locomotion [3]. Scanning mirrors provide another application of rhythmic control, generating repetitive motion to steer optical beams for confocal microscopy [4], laser micro-machining [5], and light detection and ranging (LiDAR) [6]. To achieve repeatable beam steering, the mirror should sustain oscillations with a desired scanning range and precisely controlled timing. Existing control approaches for rotational reluctance scanning mirrors combine zero-crossing synchronization with amplitude regulation based on linearized models [7]. However, these approaches do not address the joint regulation of prescribed future zero-crossing times and the scanning range while minimizing energy exchange under sparse feedback. The rhythmic nature of these systems makes neuromorphic control a promising approach, but a key challenge remains: tuning the neuronal parameters systematically to satisfy multiple control objectives.

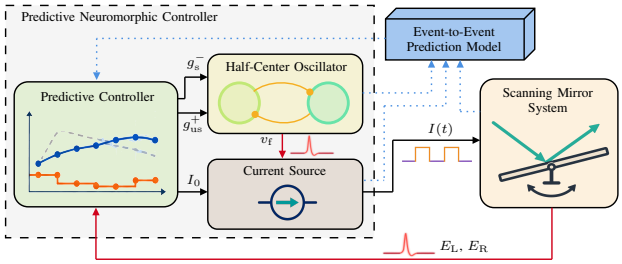


Fig. 1. Predictive Neuromorphic Control Architecture

Generalized predictive control (GPC) combines model-based prediction with online optimization over a finite horizon [8]. This principle also underlies parameterized MPC [9], in which a discrete-time model predicts the system response under a parameterized state-feedback policy and optimization selects the policy parameters that best satisfy the control objectives. A neuromorphic controller can similarly be viewed as a dynamic parameterized policy whose internal neuronal states govern actuation. Predicting the coupled evolution of the plant and controller then allows MPC to tune the controller parameters according to their anticipated effect on closed-loop behavior.

Recent developments in parameterized MPC include automatic construction of feedback policies [10] and learning-based adaptation [11]. These formulations rely on predictions of state trajectories, with performance criteria evaluated along the trajectories. They therefore do not directly exploit the information conveyed by sparse sensing events or the temporal organization of neuromorphic actuation. Event-triggered MPC [12] and more general asynchronous MPC [13] allow aperiodic feedback updates, but representative formulations retain trajectory-based prediction models and control objectives. Closing the control loop at event times therefore does not, by itself, make sensing, prediction, or regulation event-centric.

These observations motivate the development of event-centric MPC (EC-MPC), which places events at the center of sensing, prediction, and regulation. Events provide the feedback information, while an event-to-event (E2E) prediction model predicts their future occurrences. Event timing is then regulated over a horizon defined by a fixed number of future events. This creates a natural synergy

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between MPC and neuromorphic control, combining model-based prediction and online optimization with reliable sparse actuation and event-based sensing. The contributions of this paper are threefold:

1. We propose the first EC-MPC framework, and we develop a corresponding specified architecture for a scanning mirror system. The architecture uses an E2E prediction model, with events defining the feedback information, prediction horizon, and event-timing objectives.

2. We integrate EC-MPC with a neuromorphic controller to coordinate prescribed zero-crossing times and scanning range while minimizing the energy exchange between the controller and the system.

3. We develop sampling-based algorithms that obtain suboptimal solutions of the EC-MPC optimization problem through implicit evaluation of the E2E prediction model, without requiring an explicit expression for the model.

The rest of the paper is organized as follows. Section II introduces the scanning mirror dynamics and event sensing. Section III presents the neuromorphic control architecture and its parameterization. Section IV develops the EC-MPC architecture, including the E2E prediction model and sampling-based solution algorithms. Section V presents the numerical evaluation, and Section VI concludes the paper.

Notation: The sets of real numbers and nonnegative real numbers are denoted by $\mathbb{R}$ and $\mathbb{R}_+$, respectively. The set of natural numbers including 0 is $\mathbb{N}$, and $\mathbb{I}_{a:b} := \mathbb{N} \cap [a, b]$ for $0 \leq a \leq b$. The matrix or vector p-norm is $\|\cdot\|_p$, where $\|\cdot\|$ defaults to the 2-norm. The Cartesian product of sets $\mathcal{A}_1$ and $\mathcal{A}_2$ is denoted by $\mathcal{A}_1 \times \mathcal{A}_2$. For a set $\mathcal{A}$, its n-fold Cartesian product is defined recursively by $\mathcal{A}^1 := \mathcal{A}$ and $\mathcal{A}^n := \mathcal{A}^{n-1} \times \mathcal{A}$ for $n \geq 2$. For a column vector $z \in \mathbb{R}^d$, $z[a : b] := [z_a, \ldots, z_b]^\top$ denotes the subvector from the a-th to the b-th entry. Given $\bar{x} \in \mathbb{R}^n$ and a compact set $\mathcal{X} \subset \mathbb{R}$, define $\Pi_{\mathcal{X}}(\bar{x}) := \arg\min_{x \in \mathcal{X}} \|x - \bar{x}\|$. The n-fold vertical stack of z is defined recursively by $\mathtt{vec}_1\{z\} := z$ and $\mathtt{vec}_n\{z\} := [(\mathtt{vec}_{n-1}\{z\})^\top, z^\top]^\top$ for $n \geq 2$. The notation $\mathtt{diag}(z)$ denotes the diagonal matrix with diagonal entries $z_1, \ldots, z_d$. For a finite nonempty index set $\mathcal{I} = \{i_1, \ldots, i_m\}$ with a fixed ordering and a family of column vectors $\{z_i\}_{i \in \mathcal{I}}$, we define the vertical stacking operator by $\mathtt{vec}\{z_i\}_{i \in \mathcal{I}} := [z_{i_1}^\top, \ldots, z_{i_m}^\top]^\top$. For a statement P, the indicator $\mathbb{1}_{[P]}$ equals 1 if P is true and 0 otherwise. The sign function $\mathrm{sgn} : \mathbb{R} \to \{-1, 0, 1\}$ is defined by $\mathrm{sgn}(x) = -1$ for $x < 0$, $\mathrm{sgn}(x) = 0$ for $x = 0$, and $\mathrm{sgn}(x) = 1$ for $x > 0$.

II. System Modeling and Event Sensing

This section introduces the scanning mirror model and the information available through event sensing. The mechanical dynamics and reluctance actuation mechanism are first described. The photodetector events are then defined.

A. Mechanical Dynamics and Reluctance Actuation

Consider a single-axis scanning mirror attached to a ferromagnetic mover suspended by a torsional flexure (see Fig. 2). The moving assembly, hereafter referred to as the mover, is modeled as a symmetric rigid body, with the sensing tag installed on its longitudinal centerline, which passes through its center of gravity. The mover rotates about a fixed axis (the X-axis) that passes through its center of gravity, while the flexure constrains all other degrees of freedom. The mover is assumed to be balanced about the rotation axis, making gravitational torque negligible. Within the considered scanning range, the flexure is modeled as a linear torsional spring and mechanical losses are represented by viscous damping [7]. The motion is governed by the following Ordinary Differential Equation (ODE):

$$\mathcal{M}_{\mathrm{m}}: \quad J\ddot{\alpha} + b\dot{\alpha} + k_\theta \alpha = \tau(\alpha, I), \tag{1}$$

where $\alpha \in [-\alpha_{\max}, \alpha_{\max}]$ is the angular displacement from the zero-current equilibrium position, and $\alpha_{\max} > 0$ is the maximum allowable deflection angle. The constants $J > 0$, $b > 0$, and $k_\theta > 0$ denote the total moment of inertia of the moving assembly, the viscous damping coefficient, and the torsional spring stiffness, respectively. The reluctance torque $\tau(\alpha, I)$ acting on the moving assembly is approximated by

$$\tau(\alpha, I) = -\kappa \sin(\alpha) I^2, \tag{2}$$

where $\kappa > 0$ is a fitted reluctance torque coefficient. A detailed model of the reluctance force that generates this torque is provided in [7]. The coil current $I \in [0, I_{\max}]$ serves as the control input, where $I_{\max} > 0$ is the maximum allowable current. It is assumed that the current driver is sufficiently fast such that electrical transients can be neglected.

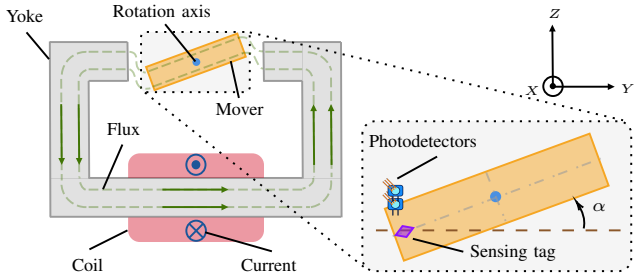


Fig. 2. 2D schematic of a resonant rotational reluctance actuator for a single-axis scanning mirror. The coil current I generates magnetic flux along the green dashed paths through the yoke, producing torque on the ferromagnetic mover about the X-axis. The inset illustrates the angular displacement α relative to the equilibrium orientation (horizontal dashed line). Two vertically separated photodetectors near equilibrium detect the sensing tag's passage, enabling zero-crossing detection and crossing-speed estimation. Circled dots and crosses indicate current flowing out of and into the drawing plane, respectively.

Appropriately timed sparse control inputs can sustain the desired motion while limiting actuator activity and energy consumption. In resonant systems, the natural exchange between kinetic and potential energy supports motion between actuation intervals. Control inputs can therefore be concentrated at suitable phases of the oscillation to compensate for dissipation and regulate its amplitude and timing. To realize such *sparse actuation*, the coil is driven by finite-duration current pulses, with *zero* current between pulses. According to (1) and (2), the current modulates the effective *angle-dependent* restoring stiffness $k_\theta + \kappa \frac{\sin(\alpha)}{\alpha} I^2$. Since the

torque depends on the squared current, reversing the current does not reverse the torque. Besides, the reluctance torque always acts toward the equilibrium position ($\alpha = 0$), and its contribution to the mechanical energy depends on the phase of the mirror motion. Specifically, the mechanical energy

$$E = \frac{1}{2}J\dot{\alpha}^2 + \frac{1}{2}k_\theta \alpha^2 \tag{3}$$

satisfies

$$\dot{E} = -b\dot{\alpha}^2 - \kappa I^2 \alpha\dot{\alpha}. \tag{4}$$

Therefore, the reluctance torque *supplies* mechanical power when the mirror moves toward the equilibrium position, i.e., $\alpha\dot{\alpha} < 0$, and *absorbs* mechanical power when the mirror moves away from it, i.e., $\alpha\dot{\alpha} > 0$. Sustained oscillations require appropriately timed current pulses to compensate for the energy dissipated by damping.

B. Sparse Sensing and Estimation

The timing of characteristic crossings provides information about the motion, allowing feedback control with limited measurement data. Detecting these crossings requires only threshold detection and time stamping, reducing reliance on sensors that continuously resolve position or velocity. This can simplify sensor hardware and reduce data acquisition, communication, and processing demands.

To enable *sparse sensing*, continuous measurements of angular position and velocity are replaced by crossing signals from two photodetectors. For a fixed incident beam under a planar scanning geometry, the angular deflection of the reflected beam from its nominal direction is 2α. The photodetectors, labeled by L and R, are positioned at beam directions corresponding to $\alpha = \gamma_\mathrm{s}$ and $\alpha = -\gamma_\mathrm{s}$, respectively, where $\gamma_\mathrm{s} > 0$ is chosen close to zero. This sensing configuration supports oscillations at different reference amplitudes satisfying $\gamma_\mathrm{s} < \alpha_\mathrm{ref} < \alpha_\mathrm{max}$. Together, the two photodetectors enable estimation of the zero-crossing time instants and the corresponding angular velocity from the order and timing of their activations. For convenience, we define the photodetector set using their labels as $\mathcal{P} = \{\mathrm{L}, \mathrm{R}\}$. Each activation of a photodetector $p \in \mathcal{P}$ is an event denoted by E_p. Each event reports the photodetector identity and crossing time, without directly reporting the direction of motion. Let $\Delta_\mathrm{det} > 0$ denote the maximum admissible time between *paired* photodetector events. A zero-crossing event E_M is inferred when two consecutive events in the combined photodetector event sequence originate from different detectors, either E_L followed by E_R or E_R followed by E_L, and their timestamps t^- and t^+ satisfy $0 < t^+ - t^- \leq \Delta_\mathrm{det}$.

III. Neuromorphic Control Architecture

This section presents the neuromorphic controller and its coupling to the actuator. The Half-Center Oscillator (HCO) dynamics [1] are first introduced, followed by the rule that converts neuronal activity into coil-current pulses. The tunable controller parameters and the tuning mechanism based on zero-crossing events are then specified.

A. Half-Center Oscillator

To generate rhythmic outputs while maintaining a simple controller architecture, we use a neuromorphic controller comprising two spiking neurons that form a half-center oscillator (HCO), as shown in Fig. 3. The circuit dynamics of each neuron are governed by the following ODEs:

$$\mathcal{M}_\mathrm{c}: \begin{cases} \tau_\mathrm{f}\dot{\nu}_{\mathrm{f},i} = -\nu_{\mathrm{f},i} + g_\mathrm{s}^- \tanh(\nu_{\mathrm{s},i} + 0.9) \\ \qquad\qquad - g_\mathrm{us}^+ \tanh(\nu_{\mathrm{us},i} + 0.9) \\ \qquad\qquad + I_{\mathrm{syn},ij} + I_{\mathrm{ext},i}, \\ \tau_\mathrm{s}\dot{\nu}_{\mathrm{s},i} = -\nu_{\mathrm{s},i} + \nu_{\mathrm{f},i}, \\ \tau_\mathrm{us}\dot{\nu}_{\mathrm{us},i} = -\nu_{\mathrm{us},i} + \nu_{\mathrm{f},i}, \end{cases} \tag{5}$$

for $i \in \mathcal{N}_\mathrm{nc} = \{n_A, n_B\}$, where $j \in \mathcal{N}_\mathrm{nc} \setminus \{i\}$ denotes the other neuron. Equations (5) describe the fast, slow, and ultra-slow internal dynamics of each neuron. Here, $\tau_{(\cdot)}$ and $\nu_{(\cdot),i}$ denote the corresponding time constants and voltages, respectively. The term $I_{\mathrm{syn},ij}$ is the synaptic current from neuron j to neuron i, and $I_{\mathrm{ext},i}$ is the externally applied current, with a nominal value of $I_{\mathrm{ext},i} = -1$. This circuit model is adopted from the mixed-feedback model in [1], designed for rhythmic control by generating individual pulse-shaped spikes rather than bursts of short spikes.

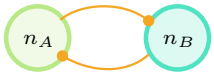


Fig. 3. A half-center oscillator comprising two spiking neurons n_A and n_B interconnected by two synapses (orange curves).

The admissible ranges of the neuron parameters g_s^- and g_us^+ are specified by $[g_\mathrm{s,min}^-, g_\mathrm{s,max}^-]$ and $[g_\mathrm{us,min}^+, g_\mathrm{us,max}^+]$, respectively. These bounds are chosen empirically to ensure that the neurons remain excitable, i.e., capable of generating spikes [1].

B. Spike-Driven Actuation

The current driver is directly coupled to one of the two neurons in the HCO, either n_A or n_B. It is activated when the fast voltage of the connected neuron reaches or exceeds a fixed threshold ν_th and remains inactive otherwise. Throughout the current paper, we use neuron n_A to activate the current driver as the default configuration. The resulting reluctance torque is given by

$$\mathcal{E}_\mathrm{act}: \quad \tau(t) = -\kappa \sin(\alpha(t))[I_0(t)]^2 \mathbb{1}_{\{\nu_{\mathrm{f},n_A}(t) \geq \nu_\mathrm{th}\}}, \tag{6}$$

where I_0 is the coil current generated by a current source.

C. Optimization-Based Parameter Tuning

The oscillation frequency and amplitude (i.e., the scanning range of the mirror) can be regulated by tuning the parameters of the excitable neurons [2] of the HCO and the coil current I_0. Specifically, g_s^- and g_us^+ influence the spike frequency and spike width, respectively. We collect these tuning parameters into the vector

$$\theta := [I_0, g_\mathrm{s}^-, g_\mathrm{us}^+]^\top. \tag{7}$$

The admissible parameter set $\Theta \subset \mathbb{R}^3$ is defined as

$$\Theta = [0, I_{\max}] \times [g^-_{\mathrm{s},\min}, g^-_{\mathrm{s},\max}] \times [g^+_{\mathrm{us},\min}, g^+_{\mathrm{us},\max}]. \quad (8)$$

Adaptive tuning of the neuron parameters has been investigated in [2]. In the current paper, we consider tuning θ through *event prediction* and *online optimization*, following the working principle of MPC. This approach is closely related to MPC with parameterized control policies, in which the parameters of a *static policy* are optimized online [9]. A distinguishing feature of our setting is that the neuromorphic controller has internal states governed by ODEs. Optimizing θ therefore amounts to tuning a *dynamic controller*. Related approaches include optimization-based tuning of proportional-integral-derivative (PID) controllers [14], whose integral action introduces an internal state.

We adopt a piecewise-constant parameterization, in which θ is held constant between consecutive occurrences of E_{M} and may be updated at each occurrence. In particular, MPC can incorporate predicted changes in references and cost weights alongside predictions of the system behavior. For $n \in \mathbb{N}$, let $t_{\mathrm{M},n}$ denote the time of the n-th occurrence of E_{M} with $t_{\mathrm{M},0} = 0$. The policy parameter θ is held constant over each half-open interval $[t_{\mathrm{M},n}, t_{\mathrm{M},n+1})$ and is given by

$$\theta_{(n)} = [I_{0,(n)}, g^-_{\mathrm{s},(n)}, g^+_{\mathrm{us},(n)}]^\top. \quad (9)$$

For each event E_{M} occurring at $t_{\mathrm{M},n} > 0$, the two associated consecutive detector events occur in the order E_{L} followed by E_{R}, or E_{R} followed by E_{L}.

Remark 1: Allowing θ to vary across intervals provides greater flexibility to accommodate time-varying control objectives and can improve control performance. Keeping θ constant over the entire prediction horizon, however, reduces the number of decision variables from $3N$ to 3.

IV. Event-Centric Model Predictive Control

This section develops the EC-MPC formulation. The E2E prediction model is first constructed, followed by the control objectives and corresponding cost function. The resulting finite-horizon optimization problem is then formulated. Finally, sampling-based algorithms for obtaining suboptimal solutions using the implicit E2E prediction model are presented, together with the online implementation.

A. Event-to-Event Prediction Model

At time $t_{\mathrm{M},n}$, we recursively predict the j-step-ahead event time instants $\hat{t}_{\mathrm{M},j|n}$ for $j \in \mathbb{I}_{0:N}$, starting from $\hat{t}_{\mathrm{M},0|n} = t_{\mathrm{M},n}$. For each $n \in \mathbb{I}_{0:N-1}$, the following E2E prediction model $F : \mathbb{R}_+ \times \Theta \to \mathbb{R}_+ \times \mathbb{R}$ determines the next event time $\hat{t}_{\mathrm{M},j+1|n}$ and the swing amplitude $\hat{A}_{[j,j+1)|n}$ over the corresponding interval $[\hat{t}_{\mathrm{M},j|n}, \hat{t}_{\mathrm{M},j+1|n})$:

$$\begin{bmatrix} \hat{t}_{\mathrm{M},j+1|n} \\ \hat{A}_{[j,j+1)|n} \end{bmatrix} = F_{\mathcal{M}_{\mathrm{m}},\mathcal{M}_{\mathrm{c}},\mathcal{E}_{\mathrm{act}}}\big(\hat{t}_{\mathrm{M},j|n}, \theta_{(j|n)}; \dot{\alpha}(\hat{t}_{\mathrm{M},j|n}), \boldsymbol{\nu}(\hat{t}_{\mathrm{M},j|n})\big), \quad (10)$$

where $\theta_{(j|n)} = [I_{0,(j|n)}, g^-_{\mathrm{s},(j|n)}, g^+_{\mathrm{us},(j|n)}]^\top$ is the virtual predicted policy parameter and

$$\boldsymbol{\nu} := \mathrm{vec}\big\{[\nu_{\mathrm{f},i}, \nu_{\mathrm{s},i}, \nu_{\mathrm{us},i}]^\top\big\}_{i \in \mathcal{N}_{\mathrm{nc}}} \quad (11)$$

is the state of the neuronal circuit. The E2E prediction model uses the torque expression (6), the mechanical dynamics (1), and the neuromorphic controller dynamics (5). At each prediction step, it is initialized with $\alpha(\hat{t}_{\mathrm{M},j|n}) = 0$, the angular velocity $\dot{\alpha}(\hat{t}_{\mathrm{M},j|n})$, and the circuit state at $\hat{t}_{\mathrm{M},j|n}$. Recursively applying the model (10) over N future occurrences of E_{M} yields the predicted event times $\{\hat{t}_{\mathrm{M},j|n}\}_{j=0}^{N}$.

A closed-form expression for $F_{\mathcal{M}_{\mathrm{m}},\mathcal{M}_{\mathrm{c}},\mathcal{E}_{\mathrm{act}}}$ is generally difficult to obtain. Instead, $[\hat{t}_{\mathrm{M},j+1|n}, \hat{A}_{[j,j+1)|n}]^\top$ can be evaluated *implicitly* through numerical simulation by integrating the coupled mirror-controller dynamics until the next occurrence of E_{M}. The dependence of the predicted event time and scanning range on the decision variable $\theta_{(j|n)}$ is therefore determined through simulation rather than an explicit expression.

B. Control Objective & Cost Function

The control objective is to regulate the mover such that the event E_{M} occurs according to a prescribed sequence of desired event times, denoted by $\{t^*_{\mathrm{M},n}\}_{n=0}^{\infty}$ with $t^*_{\mathrm{M},0} = 0$, while the swing amplitude tracks a desired sequence $\{A^*_{[n,n+1)}\}_{n=0}^{\infty}$. At the same time, the actuation effort supplied by the motor should be minimized. For convenience, we stack the decision variables as

$$\boldsymbol{\vartheta}_{N,j} = [\theta^\top_{(0|n)}, \ldots, \theta^\top_{(N-1|n)}]^\top. \quad (12)$$

Accordingly, at the event instant $t_{\mathrm{M},n}$, we consider the following finite-horizon cost function:

$$J\left(\boldsymbol{\vartheta}_N; \dot{\alpha}(\hat{t}_{\mathrm{M},j|n}), \boldsymbol{\nu}(\hat{t}_{\mathrm{M},j|n})\right) = \sum_{j=0}^{N-1} \gamma^j \Bigg[(t^*_{\mathrm{M},n+j} - \hat{t}_{\mathrm{M},j|n})^2 + w_A (A^*_{[n+j,n+j+1)} - \hat{A}_{[j,j+1)|n})^2 + w_\tau \int_{\hat{t}_{\mathrm{M},j|n}}^{\hat{t}_{\mathrm{M},j+1|n}} |\tau(t)\dot{\hat{\alpha}}(t)| \mathrm{d}t \Bigg], \quad (13)$$

where the discount factor $\gamma \in (0, 1]$ assigns progressively less weight to predictions farther into the future, $w_A \in \mathbb{R}_+$ determines the relative penalty on the amplitude-tracking error, and $w_\tau \in \mathbb{R}_+$ determines the relative penalty on the energy exchange between the controller and the mirror system. A special case of interest is periodic oscillation with a constant desired inter-event duration T and a constant desired scanning range α^*_0, such that

$$t^*_{\mathrm{M},n} = nT, \qquad |A^*_{[n,n+1)}| = A^*_0, \quad \forall n \in \mathbb{N}. \quad (14)$$

Setting $w_A = 0$ removes the amplitude-tracking objective, while setting $w_\tau = 0$ removes the actuator-effort penalty.

C. Finite-Horizon Optimization

We formulate prediction and receding-horizon optimization in the *event domain*. The event E_{M} triggers a prediction update and a new optimization. The prediction horizon spans the next N occurrences of E_{M}, whose times are predicted using an E2E prediction model. Thus, the prediction horizon is defined by a fixed number of events rather than a fixed duration. The activation of the photodetectors L and R not

only triggers MPC updates, but also provides feedback for *corrected prediction*. In particular, the short interval between consecutive activations of the two photodetectors can be used to infer the zero-crossing angular velocity. This sensing scheme follows the principle of *spike-time coding*, in which information is conveyed through the timing of discrete sensor activations.

The EC-MPC optimization problem for the scanning mirror is then formulated as

$$\begin{aligned}
\mathrm{P}_{\mathrm{EC-MPC}} : \min_{\boldsymbol{\vartheta}_N} \; & J\left(\boldsymbol{\vartheta}_N; \dot{\alpha}(\hat{t}_{\mathrm{M},0|n}), \boldsymbol{\nu}(\hat{t}_{\mathrm{M},0|n})\right) \\
\text{s.t.} \quad & \begin{bmatrix} \hat{t}_{\mathrm{M},j+1|n} \\ \hat{A}_{[j,j+1)|n} \end{bmatrix} = F_{\mathcal{M}_\mathrm{m},\mathcal{M}_\mathrm{c},\mathcal{E}_\mathrm{act}}\big(\hat{t}_{\mathrm{M},j|n}, \theta_{(j|n)}, \\
& \qquad \dot{\alpha}(\hat{t}_{\mathrm{M},j|n}), \boldsymbol{\nu}(\hat{t}_{\mathrm{M},j|n})\big), \; \forall n \in \mathbb{I}_{0:N-1}, \\
& \boldsymbol{\vartheta}_N \in \Theta^N, \\
& \hat{t}_{\mathrm{M},0|n} = t_{\mathrm{M},n}, \\
& \dot{\alpha}(\hat{t}_{\mathrm{M},0|n}) = \dot{\alpha}^*(\hat{t}_{\mathrm{M},0|n}),
\end{aligned} \tag{15}$$

where $\dot{\alpha}^*(\hat{t}_{\mathrm{M},0|n})$ is the zero-crossing angular velocity measured from the photodetector readings, the cost J is defined in (13), and $F_{\mathcal{M}_\mathrm{m},\mathcal{M}_\mathrm{c},\mathcal{E}_\mathrm{act}}$ is the E2E prediction model in (10). The optimal value of $\mathrm{P}_{\mathrm{EC-MPC}}$ is denoted by $J^\star(t_{\mathrm{M},n})$, and the corresponding set of optimal parameter sequences is denoted by $\Theta_N^\star(t_{\mathrm{M},n}) \subset \Theta^N$.

D. Sampling-Based Solution and Online Implementation

Without a closed-form expression for $F_{\mathcal{M}_\mathrm{m},\mathcal{M}_\mathrm{c},\mathcal{E}_\mathrm{act}}$, solving $\mathrm{P}_{\mathrm{EC-MPC}}$ to obtain an analytical characterization of $\Theta_N^\star(t_{\mathrm{M},n})$ and a numerical solution of an optimal sequence $\boldsymbol{\vartheta}_{N,n}^\star \in \Theta_N^\star(t_{\mathrm{M},n})$ are generally difficult. We therefore present tractable simulation-based alternatives to obtain a suboptimal parameter sequence $\tilde{\boldsymbol{\vartheta}}_{N,n}^\star$.

1) Joint dynamics rollouts: For each candidate parameter sequence, we integrate the mirror and the neuronal circuit together. Define the state of the joint system by $x = [\alpha, \omega, \boldsymbol{\nu}^\top]^\top \in \mathbb{R}^8$, where $\omega = \dot{\alpha}$ and $\boldsymbol{\nu}$ is given in (11). On the n-th predicted inter-event interval, the joint dynamics are

$$\begin{aligned}
\dot{\hat{\alpha}} &= \hat{\omega}, \\
J\dot{\hat{\omega}} &= -b\hat{\omega} - k_\theta \hat{\alpha} - \kappa \sin(\hat{\alpha}) I_{0,(j|n)}^2 \hat{\chi}, \\
\dot{\hat{\boldsymbol{\nu}}} &= f_\mathrm{c}(\hat{\boldsymbol{\nu}}; g_{\mathrm{s},(j|n)}^-, g_{\mathrm{us},(j|n)}^+), \\
\hat{\chi} &= \mathbb{1}_{\{\hat{\nu}_{\mathrm{f},n_A} \geq \nu_\mathrm{th}\}},
\end{aligned} \tag{16}$$

where f_c is the vector field in (5), including the synaptic currents. Here, $I_{0,(j|n)}$ is the commanded current amplitude; the actual coil current is $I_{0,(j|n)}\hat{\chi}$. Every candidate starts from the same state estimate

$$\hat{x}(t_{\mathrm{M},n}) = [0, \dot{\alpha}^*(t_{\mathrm{M},n}), \boldsymbol{\nu}(t_{\mathrm{M},n})^\top]^\top. \tag{17}$$

The numerical solver locates the times at which the condition $\hat{\nu}_{\mathrm{f},n_A} - \nu_\mathrm{th}$ holds, thereby determining the onset and termination of each current pulse. The integrator locates the next nontrivial zero of $\hat{\alpha}$, after the mirror has left its current crossing, and records it as $\hat{t}_{\mathrm{M},j+1|n}$. For a half-swing with one turning point, define the signed extremum by

$$\hat{A}_{[j,j+1)|n} = \sigma_{j|n} \max_{t \in [\hat{t}_{\mathrm{M},j|n}, \hat{t}_{\mathrm{M},j+1|n}]} |\hat{\alpha}(t)|, \tag{18}$$

where $\sigma_{j|n} = \mathrm{sgn}(\hat{\omega}(\hat{t}_{\mathrm{M},j|n}))$; the desired extrema use the same sign convention. Turning points are located from $\hat{\omega} = 0$. The energy-exchange term is evaluated by adding the auxiliary state q whose dynamics is given by

$$\dot{q} = |\hat{\tau}\hat{\omega}|, \qquad q(\hat{t}_{\mathrm{M},j|n}) = 0. \tag{19}$$

The terminal value of q gives the integral in (13). At each predicted crossing, only the parameter value and the auxiliary change; the mechanical velocity and all circuit states are carried forward continuously. The roll-out ends after N future crossings; thus, the prediction time depends on the candidate sequence.

Remark 2: The circuit states are internal controller variables and are retained between MPC updates. They can be read from the numerical controller, or propagated from a known initialization using the applied parameters. This does not require continuous measurements of the mechanical state.

Remark 3: A candidate is rejected if (i) the angle exceeds a predefined maximum range (i.e., $|\hat{\alpha}| > \alpha_{\max}$), (ii) a half-swing fails to reach its corresponding detector level, or (iii) no subsequent transverse zero crossing occurs within the prescribed timeout $H_{\max}$. These checks enforce the operating regime of the E2E model and limit the simulated duration of each rollout.

2) Sampling and cost-weighted refinement: We use a parameter-space search inspired by model predictive path integral (MPPI) control [15]. For event index n, at iteration r, sample perturbations around a nominal sequence $\bar{\boldsymbol{\vartheta}}_n^{(r)}$ and project onto the admissible parameter set:

$$\begin{aligned}
\boldsymbol{\vartheta}_n^{(m,r)} &= \Pi_{\Theta^N}\big(\bar{\boldsymbol{\vartheta}}_n^{(r)} + \varepsilon^{(m,r)}\big), \\
\varepsilon^{(m,r)} &\sim \mathcal{N}(0, \Sigma), \qquad m = 1, \ldots, M-1.
\end{aligned} \tag{20}$$

For $m = 0$, we set $\boldsymbol{\vartheta}_n^{(0,r)} = \bar{\boldsymbol{\vartheta}}_n^{(r)}$. Each candidate is evaluated by a joint rollout using the cost function (13). Denote its cost as $J_n^{(m,r)}$, and let $\mathcal{V}_n^{(r)}$ be the set that contains the indices of valid rollouts. For $\mathcal{V}_n^{(r)} \neq \emptyset$, define

$$\rho_n^{(r)} := \min_{m \in \mathcal{V}_n^{(r)}} J_n^{(m,r)}, \tag{21a}$$

$$w_n^{(m,r)} := \frac{\exp[-(J_n^{(m,r)} - \rho_n^{(r)})/\lambda]}{\sum_{\ell \in \mathcal{V}_n^{(r)}} \exp[-(J_n^{(\ell,r)} - \rho_n^{(r)})/\lambda]}, \quad m \in V_n^{(r)}, \tag{21b}$$

where $\lambda > 0$ is the scaling parameter that controls the sensitivity of the weights to cost variation. Smaller λ concentrates the weights on candidates with lower costs, whereas larger values distribute the weights more uniformly across valid candidates. The next nominal sequence is given by

$$\bar{\boldsymbol{\vartheta}}_n^{(r+1)} = \sum_{m \in \mathcal{V}_n^{(r)}} w_n^{(m,r)} \boldsymbol{\vartheta}_n^{(m,r)}. \tag{22}$$

By convexity of Θ, $\bar{\boldsymbol{\vartheta}}_n^{(r+1)} \in \Theta^N$. After R rounds of refinement iterations ($r = 1, 2, \ldots, R$), the valid parameter sequence with the lowest cost among all evaluated candidates, including the evaluated mean sequences, is returned as $\tilde{\boldsymbol{\vartheta}}_{N,n}^\star$.

Remark 4: This MPPI-inspired solution for the problem $\mathrm{P}_{\mathrm{EC-MPC}}$ in (15) lacks a certified optimality gap. A simpler alternative is to perform random shooting that draws a single batch uniformly from Θ^N and returns the best candidate. Enforcing $\theta_{(0|n)} = \cdots = \theta_{(N-1|n)}$ reduces the search dimension, which recovers the previously discussed constant-parameter option in Remark 1.

3) Receding-horizon implementation: At the next event $n+1$, the solution obtained for event n is shifted by one event interval with its last entry repeated to initialize the sampling, i.e.,

$$\bar{\boldsymbol{\vartheta}}^{(0)}_{n+1} = \Big[\big(\tilde{\boldsymbol{\vartheta}}^{\star}_{N,n}[4:3N]\big)^{\top}, \big(\tilde{\boldsymbol{\vartheta}}^{\star}_{N,n}[3N-2:3N]\big)^{\top}\Big]^{\top}. \quad (23)$$

After obtaining $\tilde{\boldsymbol{\vartheta}}^{\star}_{N,n}$, only $\tilde{\theta}^{\star}_{(0|n)}$ is applied; the HCO then generates the pulses until the next event update. Candidate rollouts are independent and can be evaluated in parallel. If no valid sequence is found, a defined failure action is to set the pulse amplitude to zero and retain the circuit gains; this action does not guarantee sustained oscillation.

For two detector times $t_n^- < t_n^+$ associated with the same passage, the feedback estimates are

$$t_{\mathrm{M},n} \simeq \frac{t_n^- + t_n^+}{2}, \qquad \dot{\alpha}^*(t_{\mathrm{M},n}) \simeq \frac{2\sigma_n \gamma_{\mathrm{s}}}{t_n^+ - t_n^-}, \quad (24)$$

where $\sigma_n = +1$ for E_{R} followed by E_{L}, and $\sigma_n = -1$ for the reverse order. These estimates become available only at t_n^+. The event-time formulation assumes that this short delay and the optimization time are negligible. A causal implementation with appreciable delay must retain the previous parameters until the new solution is available and propagate both the mechanical and circuit states to that application time; it cannot apply a newly computed parameter retroactively at the reconstructed crossing.

V. Numerical Simulation

This section evaluates the proposed EC-MPC framework through numerical simulations. The simulation setup is first described. Control performance is then assessed in terms of zero-crossing times, scanning-range tracking, and mechanical energy exchange. Finally, the computational cost of the simulation-based solution strategy and its implications under model mismatch are discussed.

A. Simulation Setup

We consider a nominal numerical example combining the scanning mirror in Section II with the HCO in Section III. The reference oscillation frequency is $f_{\mathrm{ref}} = 10\,\mathrm{Hz}$, corresponding to a desired inter-event duration $T = 1/(2f_{\mathrm{ref}}) = 50\,\mathrm{ms}$. The mechanical parameters are selected to be compatible with the desired reference oscillation frequency.

1) Mechanical system: Table I summarizes the mechanical parameters and operating limits. The zero-current natural frequency of the mover is chosen as $f_0 = 9.9\,\mathrm{Hz}$ to account for the additional restoring torque during current pulses. The stiffness and damping are determined by

$$k_\theta = J(2\pi f_0)^2, \qquad b = 2\zeta J(2\pi f_0), \quad (25)$$

where $J = 10^{-7}\,\mathrm{kg\,m^2}$ and $\zeta = 0.01$. The torque coefficient $\kappa = 0.0309\,\mathrm{N\,m/A^2}$ is obtained based on the small-angle coefficient $0.5385\,\mathrm{mN\,m/(deg\,A^2)}$ in [7]. The simulation retains the $\sin\alpha$ dependence in (2). The current limit is $I_{\max} = 0.12\,\mathrm{A}$ to match the mechanical stiffness.

TABLE I
Mechanical parameters and operating limits.

Parameter	Value	Parameter	Value
J	$10^{-7}\,\mathrm{kg\,m^2}$	ζ	0.01
f_0	9.9 Hz	κ	$30.854\,\mathrm{mN\,m/A^2}$
k_θ	$0.3869\,\mathrm{mN\,m/rad}$	b	$0.1244\,\mu\mathrm{N\,m\,s/rad}$
$I_{\max}$	0.12 A	$\alpha_{\max}$	10°
γ_{s}	0.05°	A_0^*	5°

2) Neuronal circuit: The two neurons are identical and connected by reciprocal *inhibitory* synapses. The synaptic current is given by [2]

$$I_{\mathrm{syn},ij} = \frac{g_{\mathrm{syn}}}{1+\exp[-2(\nu_{\mathrm{s},j}+1)]}, \quad (26)$$

where $g_{\mathrm{syn}} < 0$ is the synaptic gain. The default neuronal gains are $(g_{\mathrm{s}}^-, g_{\mathrm{us}}^+) = (2, 1.5)$, and the dimensionless baseline time constants are set as $(1, 10, 100)$. Accordingly, the simulated HCO has a recurrence dimensionless period $P_{\mathrm{H}} \simeq 286.320$ between successive upward threshold crossings corresponding to neuron n_A, and the time constants are then given by

$$c_{\mathrm{t}} = \frac{T}{P_{\mathrm{H}}} \simeq 174.63\,\mu\mathrm{s}, \quad (\tau_{\mathrm{f}}, \tau_{\mathrm{s}}, \tau_{\mathrm{us}}) = c_{\mathrm{t}}(1, 10, 100). \quad (27)$$

The nominal current pulse has a period of T, generating two pulses per mechanical period. The duration of each current pulse is approximately $13.583\,\mathrm{ms}$. The time constants remain fixed when the gains are optimized online. The parameters of the neuronal circuit are summarized in Table II.

TABLE II
Nominal neuronal parameters.

Parameter	Value	Parameter	Value
g_{syn}	-1	$I_{\mathrm{ext},i}$	-1
g_{s}^-	2	g_{us}^+	1.5
ν_{th}	0	τ_{f}	0.17463 ms
τ_{s}	1.7463 ms	τ_{us}	17.463 ms

3) Initialization and event feedback: The reference event times and signed half-swing extrema are chosen to support periodic scanning (cf. (14)):

$$t^*_{\mathrm{M},j} = jT, \qquad A^*_{[j,j+1)} = (-1)^j A_0^*, \qquad A_0^* = 5^\circ. \quad (28)$$

The system is initialized with an established oscillation at a positive-going zero crossing, with $\alpha(0) = 0$, $\dot{\alpha}(0) = 5.5137\,\mathrm{rad/s}$, and $I_0 \simeq 32.606\,\mathrm{mA}$. The initial circuit state is given by

$$\begin{aligned} \boldsymbol{\nu}_{n_A}(0) &\simeq (-0.7739, -0.4348, -0.4461), \\ \boldsymbol{\nu}_{n_B}(0) &\simeq (-2.3198, -2.4326, -2.1535). \end{aligned} \quad (29)$$

The current pulses are expected to act during the return toward the equilibrium ($\alpha = 0$), sustaining the oscillation in the presence of damping.

The photodetectors are placed at $\alpha = \pm\gamma_{\mathrm{s}}$. Opposite-detector activations are paired when their separation is below $\Delta_{\mathrm{pair}} = 0.05T = 2.5\,\mathrm{ms}$, and the feedback estimates are computed using (24). We assume noiseless timestamps and negligible sensing and computation delays, as discussed in Section IV-D. The controller uses only the event-derived mechanical estimate and its retained circuit state.

4) MPC settings: The settings for MPC are listed in Table III. The standard deviations for parameter sampling are $(\sigma_I, \sigma_{\mathrm{s}}, \sigma_{\mathrm{us}})$, which form the covariance in (20) as $\Sigma = \mathtt{diag}(\mathtt{vec}_N([\sigma_I^2, \sigma_{\mathrm{s}}^2, \sigma_{\mathrm{us}}^2]^\top))$. The horizon for MPC spans $N = 4$ half-swings, with a nominal prediction time of $NT = 0.2\,\mathrm{s}$.

TABLE III

MODEL PREDICTIVE CONTROL PARAMETERS.

Parameter	Value	Parameter	Value
N	4	M	256
R	2	γ	1
$[g^-_{\mathrm{s,min}}, g^-_{\mathrm{s,max}}]$	$[1.8, 2.2]$	$[g^+_{\mathrm{us,min}}, g^+_{\mathrm{us,max}}]$	$[1.3, 1.7]$
σ_I	$12\,\mathrm{mA}$	$(\sigma_{\mathrm{s}}, \sigma_{\mathrm{us}})$	$(0.04, 0.04)$
λ	$250\,\mathrm{ms}^2$	$H_{\max}$	$100\,\mathrm{ms}$
ϵ_{rel}	10^{-7}	ϵ_{abs}	10^{-9}
$h_{\max}$	$87.315\,\mu\mathrm{s}$	K	200
s_{rng}	1	Δ_{pair}	$2.5\,\mathrm{ms}$

The nominal kinetic energy of the mover system is given by $E_0 = \frac{1}{2}k_\theta(\alpha_0^*)^2 \simeq 1.4733\,\mu\mathrm{J}$, and the cost weights are then chosen as

$$w_A = \frac{T^2}{(\alpha_0^*)^2} \simeq 0.3283\,\mathrm{s}^2/\mathrm{rad}^2, \qquad w_\tau = 0.05\frac{T^2}{E_0} \simeq 84.843\,\mathrm{s}^2/\mathrm{J}. \tag{30}$$

The adaptive fourth-order Runge–Kutta method is used for numerical rollouts with relative tolerance ϵ_{rel}, scaled absolute tolerance ϵ_{abs}, and maximum step $h_{\max} = \tau_{\mathrm{f}}/2$. The plant and prediction model use the same nominal parameters. The number of simulated inter-event intervals is $K = 50$, corresponding to approximately $2.5\,\mathrm{s}$.

B. Results and Performance Evaluation

To be added: figures and their accompanying discussion

C. Discussion

To be added: the discussion of limitations of the current approach, model mismatch, constraint satisfaction, and computational cost.

VI. Conclusions and Future Work

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